Dynamic Pathfinding Agent Using Informed Search

1. Objective

The objective of this project is to design a Dynamic Pathfinding Agent that navigates a grid-based environment using informed search algorithms. The agent must adapt to randomly appearing obstacles by detecting path obstructions and re-planning in real time.

2. Environment Description

The environment is a 2D grid consisting of free cells and obstacles.

Features:

- User-defined grid size (rows × columns)
- Fixed start and goal nodes
- Random map generation with configurable obstacle density
- Interactive wall placement/removal using mouse input
- No static map files are used

3. Algorithms Implemented

Greedy Best-First Search (GBFS)

GBFS selects nodes based only on heuristic distance to the goal.

$f(n) = h(n)$

- Fast execution
- Not guaranteed to find optimal paths

A* Search

A* evaluates nodes using both path cost and heuristic estimate.

$f(n) = g(n) + h(n)$

- Complete and optimal
- Higher computational cost than GBFS

4. Heuristic Functions

Two heuristics are implemented and selectable via GUI:

- Manhattan Distance – suitable for grid movement
- Euclidean Distance – provides straight-line estimation

5. Dynamic Obstacles and Re-planning

During agent movement, obstacles spawn randomly with a small probability.
If a new obstacle blocks the remaining planned path:

- The agent detects the obstruction
- Re-planning is performed from the agent's current position
- Re-search is triggered only when necessary, improving efficiency

6. Visualization and Metrics

A graphical interface is implemented using Tkinter.

Visual Elements:

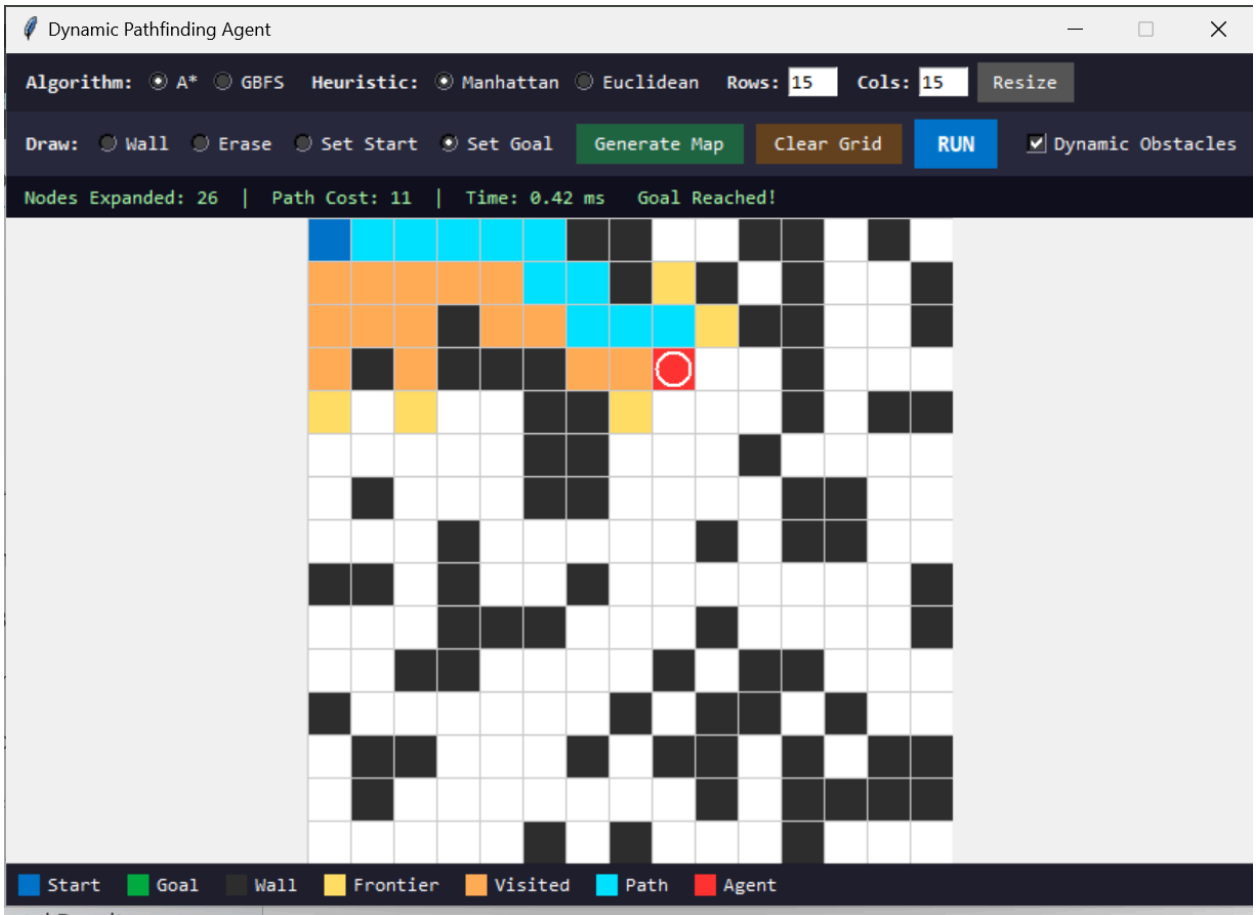
- Start, goal, walls, visited nodes, frontier nodes, final path, and moving agent

Metrics Displayed:

- Number of expanded nodes
- Path cost
- Execution time (milliseconds)

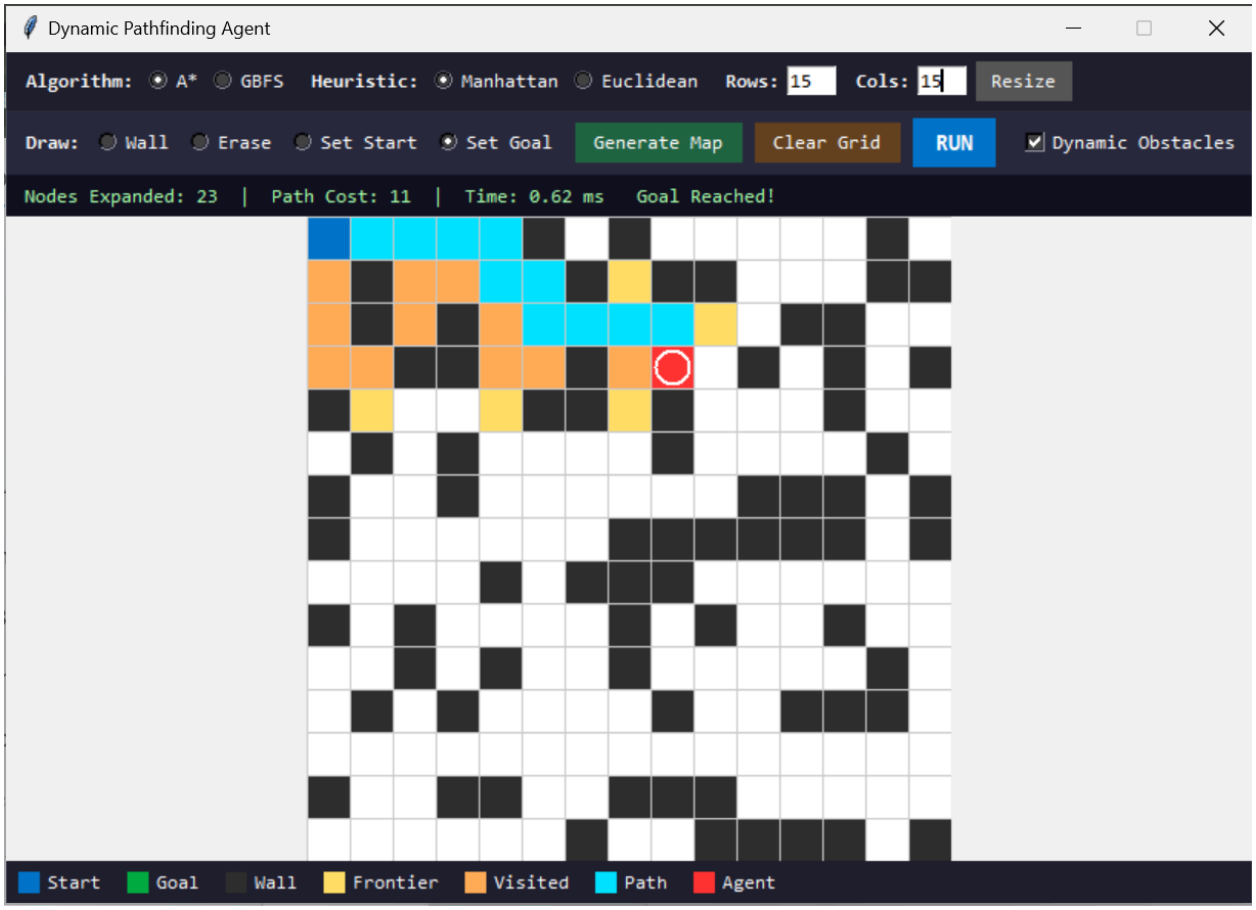
7. Experimental Results

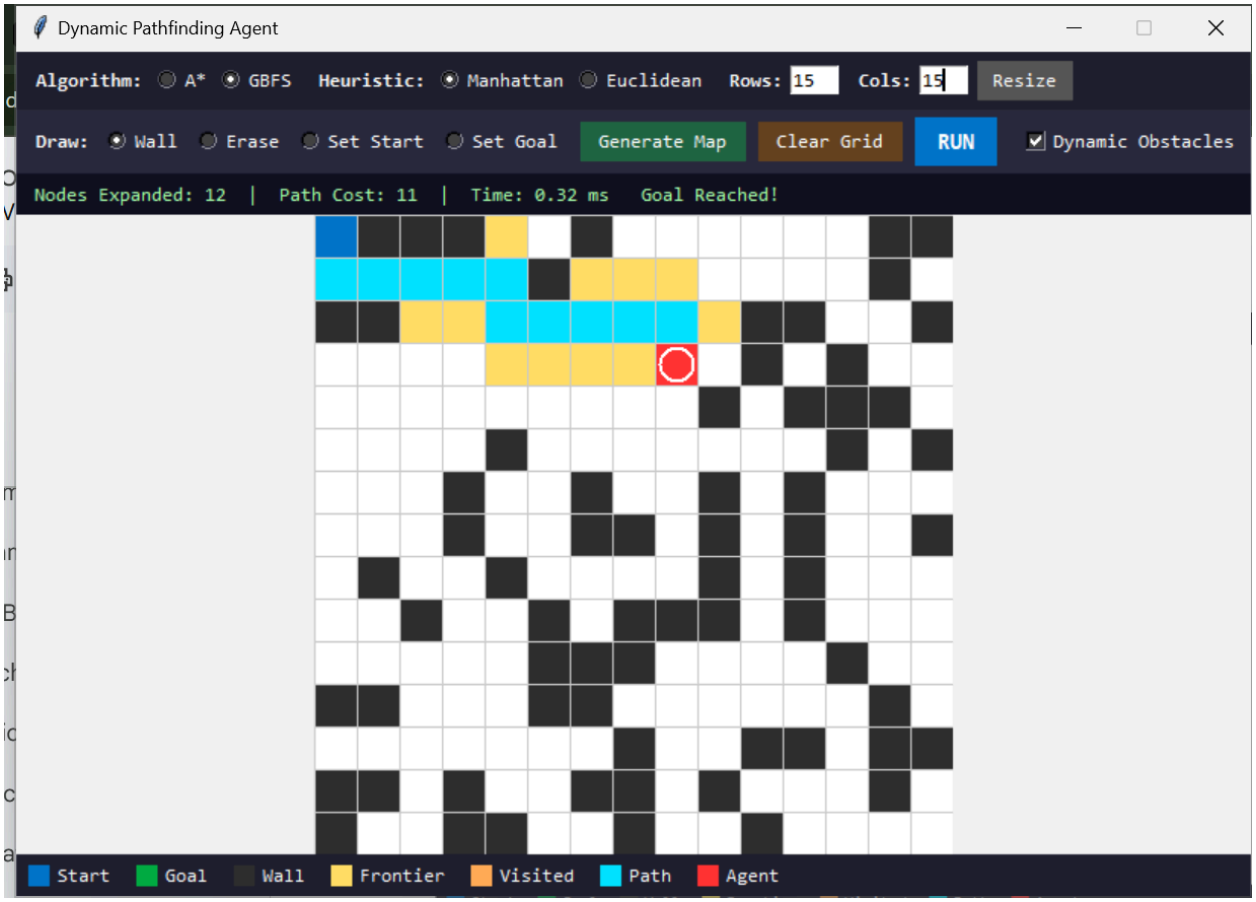
- Best Case: Sparse obstacles; both algorithms perform efficiently
- Worst Case: Dense obstacles; A* performs better due to optimality, while GBFS explores more nodes

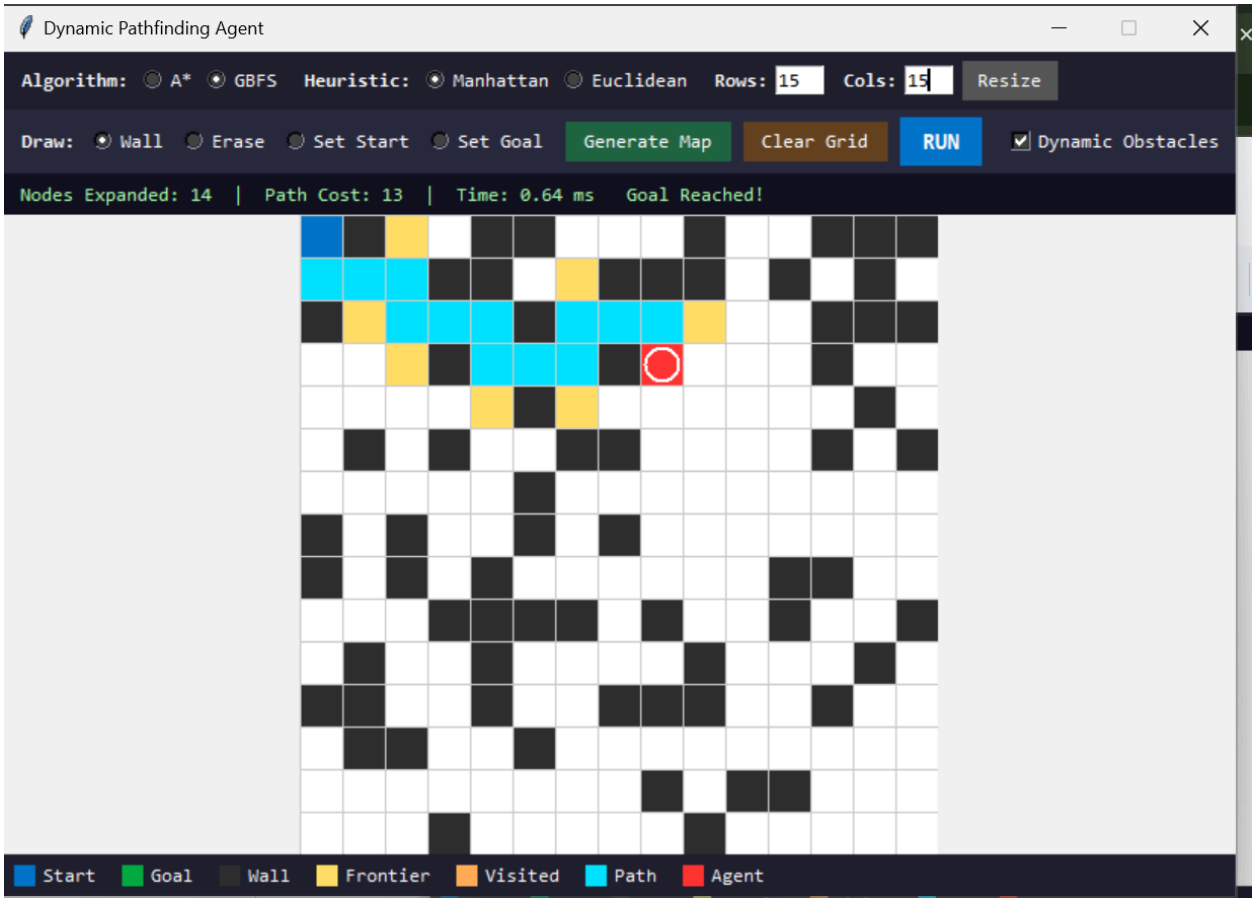


ental Results

8. Conclusion


Dynamic Pathfinding Agent
Algorithm: A* GBFS
Heuristic: Manhattan Euclidean
Rows: 15
Cols: 15
Resize
Draw: Wall Erase Set Start Set Goal
Generate Map
Clear Grid
RUN
Dynamic Obstacles
Nodes Expanded: 23 | Path Cost: 11 | Time: 0.62 ms Goal Reached!
Start
Goal
Wall
Frontier
Visited
Path
Agent


Dynamic Pathfinding Agent
Algorithm: A* GBFS
Heuristic: Manhattan Euclidean
Rows: 15
Cols: 15
Resize
Draw: Wall Erase Set Start Set Goal
Generate Map
Clear Grid
RUN
Dynamic Obstacles
Nodes Expanded: 12 | Path Cost: 11 | Time: 0.32 ms Goal Reached!
Start
Goal
Wall
Frontier
Visited
Path
Agent



8. Conclusion

This project demonstrates effective use of informed search in dynamic environments. A* guarantees optimal solutions, while GBFS offers faster performance with reduced reliability. The real-time re-planning mechanism enables robust navigation in changing environments.